MDT × GNN research ledger

This ledger turns the GNN continuation into falsifiable tests. Labels are never used for fitting or model selection; they are read only after a run to compute AMI. Every claim is evaluated over the same data splits and seeds.

Why a GNN is worth one controlled pass

The previous MLP study already closes any claim that a neural factorisation of a fixed MDT Gram matrix can exceed its truncated SVD: at the global optimum it is the same rank-k factorisation. A GNN does not change that theorem. It can, however, change two questions that the theorem does not answer:

1. Does message passing give a better inductive map for unseen samples than a feature-only encoder and Nyström?
2. Does a graph-native self-supervised objective learn a more cluster-useful geometry than fixed MDT consensus, without using labels?

The suite in `experiments/gnn_mdt/closure.py` tests both questions and isolates multi-view fusion from extra model capacity.

Hypotheses and closure rules

ID	hypothesis	decisive comparisons	continue only if
G1	A GNN distils fixed MDT more faithfully than an MLP.	`teacher_gnn` vs `teacher_mlp`; Procrustes fidelity and train AMI.	Fidelity improves materially and is stable over 5 seeds. This is an engineering result, not an accuracy result.
G2	Graph message passing improves MDT out-of-sample extension.	`teacher_gnn` vs `mdt_nystrom` and `teacher_mlp`; test and inductive AMI.	Mean paired gain ≥0.02 and the gain is not driven by one dataset/seed.
G3	A graph autoencoder escapes the fixed-SVD ceiling usefully.	`gae_uniform` vs `mdt_consensus`, `features`, and `gae_mlp`.	Mean paired train/test gain ≥0.02 and wins on at least 4/6 datasets.
G4	Infomax, rather than edge reconstruction, is the missing objective.	`dgi_uniform` vs `mdt_consensus` and `gae_uniform`.	Same threshold as G3; otherwise close this objective.
G5	Learning view importance solves heterogeneous-view dilution.	learned vs uniform variants on Wikipedia/Caltech plus all datasets.	≥0.02 mean gain, stable attention weights, and no regression on homogeneous datasets.
G6	GNN computation is preferable to sparse truncated SVD.	graph build + training + inference vs graph build + `svds`; report both.	End-to-end time wins, or a measured repeated-query break-even point is realistic.
A2a	AlphaFold-style recycling improves the parametric OOS map.	`teacher_mlp_recycle` vs `teacher_mlp`, inductive AMI primary.	≥0.02 mean paired gain with the interval excluding zero, and the extra inference cost justified.
A2b	An AlphaFold-style pair target beats coordinate regression.	`teacher_mlp_pair` vs `teacher_mlp`; interaction read from `teacher_mlp_pair_recycle`.	Same threshold; a fidelity loss is acceptable only if inductive AMI clears it.

The automated verdict uses paired effects and a 95% interval clustered by dataset: the five seeds are averaged first, so they are not treated as 30 independent benchmark tasks. Continue when the mean clears the threshold and the interval excludes zero; stop when the interval excludes the minimum useful effect; otherwise remain inconclusive. It refuses to issue a verdict from fewer than five datasets. A paper-facing positive result still needs the locked configuration and evaluation on untouched data.

Fairness constraints

- Identical train/test splits, per-view standardisation, kNN graphs, embedding dimension, and KMeans protocol for all methods.
- `teacher_*` uses the fixed MDT SVD as its only target. `gae_*` and `dgi_*` never see the SVD target or labels.
- `*_mlp` removes message passing while retaining comparable projection capacity. `*_uniform` freezes relation weights; `*_learned` changes only those weights.
- Nyström is the OOS baseline; sparse truncated `svds` is the compute baseline; MDT consensus is the strongest surviving unsupervised geometry baseline.
- Report both test-only KMeans AMI (compatible with the existing paper) and the stricter inductive AMI obtained by assigning test points to train-fitted centroids.

Method: the two distillation students, `teacher_gnn` and `teacher_mlp`

Both variants are parametric inductive maps. They learn a function from raw multi-view features to the MDT embedding, so that an unseen sample can be embedded by a forward pass instead of by redoing the spectral decomposition. They are trained on exactly the same target with exactly the same architecture, and differ in one line of code. Everything below is implemented in `experiments/gnn_mdt/closure.py`.

The teacher they distil

The target is built once per (dataset, seed), before any network exists.

1. **Graphs.** For each view `v`, standardise the features, take a Gaussian kernel over the `k = floor(log n_train)` nearest neighbours with a global median bandwidth estimated on at most 200 points, symmetrise with `max(W, Wᵀ)`, and row-normalise to get a stochastic matrix `P_v` (`build_graphs`, `knn_transitions`, `:107-148`). The self-neighbour is included, so `P_v` has non-zero diagonal. One graph per view, same node set.
2. **Trajectory.** Draw `T = 4` weight vectors from a Dirichlet(1, …, 1) over the views, seeded by the run seed (`trajectory`, `:151-152`). Row `t` gives the mixing coefficients `α_t,v` for diffusion step `t`.
3. **Operator.** Compose the steps: `W = (Σ_v α_T,v P_v) ⋯ (Σ_v α_1,v P_v)` (`mdt_operator`, `:155-160`). This is a single MDT trajectory, not the 8-trajectory consensus used by `mdt_consensus`.
4. **Embedding.** Sparse truncated SVD of `W`, keep components 1…k of the left factor scaled by their singular values, discarding component 0 as the trivial stationary direction (`svd_embedding`, `:177-185`). `k` equals the number of classes, which is the only

place class information enters, and it enters as a dimension count rather than as labels.

The result, `svd_train` (`n_train × k`), is the sole supervision signal for both students (`:461`). Before the loss it is standardised per column (`:335`), so every retained direction carries equal weight and the students are not asked to reproduce the spectral decay. Fidelity is later measured after the same standardisation, so the two are consistent.

Shared architecture

`MultiViewSAGE` (`:229-287`), a two-hop relation-aware GraphSAGE:

stage	operation	shape
per-view input	`h0_v = relu(W_v x_v + b_v)`, one `Linear` per view	`n × 128`
view fusion	`h0 = Σ_v α_v h0_v`, `α = softmax(fusion_logits)`	`n × 128`
hop 1	`h1 = relu(layer1[ h0 ‖ Σ_v α_v (P_v h0_v) ])`, `layer1: 256 → 128`	`n × 128`
hop 2	`z = layer2[ h1 ‖ Σ_v α_v (P_v h1) ]`, `layer2: 256 → k`, no activation	`n × k`

Three details matter. Hop 1 diffuses each view's own `h0_v` on that view's own graph before fusing, whereas hop 2 diffuses the single shared `h1` on every graph (`h1_views = [h1 for _ in graphs]`, `:263`) — the per-view identity of the representation survives only one hop. The concatenation keeps an untransformed copy of the node's own state alongside the neighbourhood average, which is what separates GraphSAGE from GCN, where the two are summed. And for the two teacher variants `fusion_logits` is frozen (`requires_grad` is set only for `*_learned`, `:239-241`), so fusion is uniform `1/V`; learned fusion is tested separately under G5.

For MSRC-v5 (5 views of 24/576/512/256/254 dimensions, `k = 7`) this is 242,951 trained parameters: 208,256 in the input projections, 32,896 in `layer1`, 1,799 in `layer2`. The `Bilinear` discriminator (`:242`) exists on the module but is used only by the DGI objective; under the teacher loss it receives no gradient.

The single difference

`_aggregate` returns a zero tensor when `message_passing` is false (`:251-256`):

```
def _aggregate(self, graphs, values):
    if not self.message_passing:
        return torch.zeros_like(values[0])
    return self._fuse([torch.sparse.mm(graph, value)
                       for graph, value in zip(graphs, values)])
```

So `teacher_mlp` computes `h1 = relu(layer1[ h0 ‖ 0 ])` and `z = layer2[ h1 ‖ 0 ]`. The right half of both weight matrices still exists and still receives weight decay; it simply multiplies zeros. Parameter count, optimiser, learning rate, epoch budget, patience, seed, splits, graphs and evaluation protocol are identical. The variant is selected by a suffix test, `message_passing = not variant.endswith("mlp")` (`:325`). This is what makes the contrast attributable to message passing rather than to capacity — a comparison against a smaller or differently shaped MLP would confound the two.

The match is on nominal parameters, not on effective capacity: weights that multiply zeros compute no function, so a GNN win over this control is still open to the reading "the extra live parameters helped". Matching effective capacity while removing message passing is impossible, because the hop is the capacity under test. Amendment A3 closes that gap from the other side, with a control that keeps every weight live and destroys only the neighbourhoods.

Training

Full-batch, no minibatching, no validation split, no labels (`fit_model`, `:318-385`):

- loss `MSE(z, teacher)` against the standardised target;
- Adam, learning rate 3e-3, weight decay 1e-4;
- gradient-norm clipping at 5.0;
- up to 100 epochs, early stopping after 40 epochs without a new best training loss, then the best state is restored before inference.

Early stopping on training loss rather than on a held-out score is deliberate: a held-out selection signal would need labels or a second target, and the ledger's rule is that nothing is selected on labels. The cost is that the epoch budget is not tuned per dataset, which is one reason the timings in G6 are not a tuned-implementation benchmark.

Out-of-sample inference

Test points are never in the training graph. They are attached by a directed row-normalised bipartite graph `B_v` of shape `n_test × n_train`, built with the same bandwidth and the same `k` (`:128-134`); test points connect only to training points, never to each other, and each test row is independent of the rest of the test set.

`forward_test` (`:267-287`) then mirrors the training forward pass with the train-side states as the message source:

```
h0_test  = Σ_v α_v relu(W_v x_test,v)
h1_test  = relu(layer1[ h0_test ‖ Σ_v α_v (B_v h0_train,v) ])
z_test   = layer2[ h1_test ‖ Σ_v α_v (B_v h1_train) ]
```

For `teacher_mlp` both aggregation terms are zero, so the map collapses to `z_test = f(x_test)`: a pure feature-to-embedding function that needs neither the graph nor the training set at query time. This asymmetry is the substantive difference between the two at deployment, and it is why the G6 break-even question is well posed for the MLP and not for the GNN.

Both are compared against `mdt_nystrom`, which uses the same `B_v` but no learned parameters: it replays the MDT trajectory with the final step replaced by the bipartite operator and projects onto the teacher's right singular vectors (`mdt_oos_operator`, `:163-174`; `:425`).

What is recorded per run

`fidelity` (Procrustes agreement with the teacher on train), `train_ami`, `test_ami` (KMeans refit on the test embedding), `inductive_ami` (test points assigned to train-fitted centroids), `graph_seconds`, `train_seconds`, `inference_seconds`, the converged loss, and the softmax fusion weights. Labels touch none of this until AMI is computed after the run.

Commands

```
python -m unittest tests.test_gnn_mdt
python -m experiments.gnn_mdt.closure --smoke
python -m experiments.gnn_mdt.closure --config experiments/gnn_mdt/config.yml
python -m experiments.gnn_mdt.closure --datasets MSRC-v5 Handwritten --seeds 0 1 --epochs 100 --
output /tmp/gnn-pilot.jsonl
python -m experiments.gnn_mdt.closure --config experiments/gnn_mdt/config.yml --resume
python -m experiments.gnn_mdt.closure --summarize results/gnn_mdt/metrics.jsonl
python -m experiments.gnn_mdt.closure --summarize results/gnn_mdt/metrics.jsonl --summary-output
results/gnn_mdt/summary.json
python -m experiments.gnn_mdt.closure --summarize results/gnn_mdt/validation_metrics.jsonl --summary-
output results/gnn_mdt/validation_summary.json
python -m experiments.gnn_mdt.closure --variants teacher_mlp teacher_mlp_recycle teacher_mlp_pair
teacher_mlp_pair_recycle --output results/gnn_mdt/af_metrics.jsonl --summary-output results/gnn_mdt/
af_summary.json
python -m experiments.gnn_mdt.closure --output results/gnn_mdt/control_metrics.jsonl --summary-output
results/gnn_mdt/control_summary.json
python -m experiments.gnn_mdt.closure --datasets MSRC-v5 Handwritten Wikipedia UCI OutdoorScene
Caltech101-7 3Sources BBCSport Prokaryotic Reuters-1200 NUS-WIDE MNIST-4 --output results/gnn_mdt/
power_metrics.jsonl --summary-output results/gnn_mdt/power_summary.json
```

Do not read `results/gnn_mdt/summary.json` as the screen result. It holds the validation numbers — `mdt_nystrom` test AMI mean 0.2495, which is the validation six, against 0.6026 on `metrics.jsonl` — and it is equal to `validation_summary.json` on every field. The `teacher_gnn` − `teacher_mlp` test effect reads -0.0215 there and +0.0118 on the screen grid, so the two disagree on sign. Always recompute with `--summarize` against the file you mean.

The full configuration is intentionally five-seed and six-dataset. During development, reduce `epochs`, the seed list, and the dataset list in a copied configuration; do not use those reduced runs for a verdict.

Locked run: 6 datasets × 5 seeds

Run completed with the checked-in configuration (800 training and 400 test samples maximum, 100 epochs, embedding dimension equal to class count). The raw 300 rows are in `results/gnn_mdt/metrics.jsonl`. Intervals below are paired 95% t intervals across the six dataset means (five seeds are averaged within each dataset).

question	paired AMI effect	evidence	verdict
GNN teacher vs Nyström, test	+0.0359 [0.0153, 0.0564]	wins 25/30; inductive effect +0.0349	**continue to untouched-data validation**
GNN teacher vs MLP teacher, test	+0.0118 [-0.0097, 0.0333]	inductive effect +0.0195 [-0.0003, 0.0392]	**inconclusive / graph-specific gain not established**
GNN vs MLP teacher fidelity	-0.0009 [-0.0036, 0.0017]	mean fidelity 0.9895 vs 0.9904	**dead end as a fidelity claim**
GAE message passing vs GAE-MLP, test	+0.0402 [0.0258, 0.0545]	wins 26/30	**continue: the graph is useful**
learned GAE vs MDT consensus, test	+0.0215 [-0.0412, 0.0841]	wins 19/30; −0.080 on Wikipedia	**inconclusive / not a general replacement**
learned vs uniform GAE fusion, train	+0.0008 [-0.0069, 0.0085]	mean attention deviation from uniform only 0.0071	**dead end**
DGI vs MDT consensus, train	-0.3011 [-0.3988, -0.2033]	loses 30/30 although its loss converges	**dead end: objective misalignment**
GNN teacher compute vs sparse SVD	12.1× slower training	2.535 s vs 0.209 s mean, excluding shared graph build	**dead end for compute**

The screen left two provisional survivors: parametric SVD distillation for OOS and graph-autoencoder message passing. Their required untouched-data validation is reported below.

Untouched validation and final closure

The locked configuration was evaluated without tuning on six additional datasets: 100Leaves, Yale, 3Sources, BBCSport, Cora, and CiteSeer. Five seeds produced 240 complete, unique, finite result rows in `results/gnn_mdt/validation_metrics.jsonl`. Cora and CiteSeer have almost no recoverable cluster signal for any evaluated method; they are retained in the primary analysis rather than removed post hoc.

question	validation paired AMI effect	replication result	final decision
teacher GNN vs Nyström, test	+0.0602 [-0.0454, 0.1659]	22/30 wins, but the interval crosses zero and the largest gain is on 100Leaves	**close as a general GNN claim**
teacher GNN vs MLP, test	-0.0215 [-0.0872, 0.0442]	14/30 wins; inductive +0.0042 [-0.0308, 0.0393]	**dead end: message passing is not the source of OOS gain**
GNN vs MLP teacher fidelity	+0.0065 [-0.0103, 0.0233]	no replication of a fidelity advantage	**dead end**
GAE message passing vs GAE-MLP, test	+0.0107 [-0.0803, 0.1018]	18/30 wins and strong dataset heterogeneity	**close: screen gain did not replicate**
learned GAE vs MDT consensus, test	-0.0099 [-0.0719, 0.0520]	12/30 wins	**dead end as an MDT replacement**
learned vs uniform GAE fusion, train	+0.0058 [-0.0069, 0.0185]	below the +0.02 minimum useful effect	**dead end, replicated**
GNN teacher compute vs sparse SVD	9.8× slower training	excludes shared graph construction	**dead end, replicated**

Pooling the screen and validation as a secondary 12-dataset analysis clarifies the only apparent positive result. Teacher distillation beats Nyström by +0.0480 test AMI [0.0034, 0.0927], but the GNN-minus-MLP effect is -0.0049 [-0.0352, 0.0254]. Thus **parametric OOS extension remains worth studying, but it is an MLP/DDM result rather than evidence for a GNN**. The pooled GAE message-passing effect is +0.0254 [-0.0134, 0.0643], so it also fails the stability rule.

All launched GNN hypotheses are therefore closed under this formulation:

- G1 fidelity: dead end.
- G2 graph-specific OOS extension: dead end; retain feature-only parametric OOS.
- G3 graph autoencoder as an MDT replacement: dead end.
- G4 DGI: dead end from objective misalignment.
- G5 learned global view weights: dead end.
- G6 compute advantage: dead end.

Reopening a GNN branch now requires a materially different hypothesis, such as sample-dependent view gating or genuinely relational side information. More seeds, hidden-width sweeps, or another global fusion scalar do not address the observed failure modes.

Detailed records: G1, G2, G6

The three hypotheses below are expanded because they carry the load-bearing negative results. Every number is recomputed from the two checked-in result files with the same estimator the automated summary uses: the five seeds are averaged within a dataset first, then a paired 95% Student interval is taken across the dataset means. Win counts remain seed-level and are reported only as a consistency diagnostic, never as evidence on their own.

G1 — a GNN distils fixed MDT more faithfully than an MLP

Methodology. Both `teacher_gnn` and `teacher_mlp` regress one identical target: the rank-k factor `U_k Σ_k` of a single fixed MDT operator, obtained by sparse truncated SVD of the train graphs and passed as the only supervision signal (`run_one`, `closure.py:420-422` and `:461`). The target is standardised per column before the loss, so the two variants optimise the same objective on the same scale. They also share the whole architecture in `MultiViewSAGE`: per-view input projections to 128 hidden units, a softmax fusion over views, and two `Linear(2·hidden, ·)` layers. The single difference is that `_aggregate` returns zeros for the `*_mlp` variant (`closure.py:253-254`), so the neighbour half of each concatenation is dead while the parameter count, the optimiser, the epoch budget and the early-stopping patience are unchanged. That matches nominal parameters but not effective capacity, since dead weights compute no function; the residual gap is closed by the shuffled-graph control in A3, whose teacher arm agrees with the verdict below. Fidelity is Procrustes agreement with the teacher after per-column standardisation of both matrices, `1 - ||AR - B||²_F / ||B||²_F` (`procrustes_fidelity`, `closure.py:403-408`); it equals 1 when the student recovers the teacher up to rotation. `mdt_nystrom` is assigned fidelity 1.0 by definition because it is the teacher, and it must therefore be excluded from this comparison.

Evidence. Per-dataset means over five seeds.

dataset	classes	fidelity GNN	fidelity MLP	Δ fidelity	Δ train AMI
MSRC-v5	7	0.9991	0.9999	-0.0008	+0.0045
Handwritten	10	0.9936	0.9935	+0.0001	+0.0024
Wikipedia	10	0.9732	0.9701	+0.0031	-0.0095
UCI	10	0.9905	0.9920	-0.0015	-0.0022
OutdoorScene	8	0.9837	0.9883	-0.0046	+0.0082
Caltech101-7	7	0.9970	0.9989	-0.0019	+0.0078
100Leaves	100	0.5959	0.5567	+0.0392	+0.0066
Yale	15	0.9997	0.9999	-0.0002	-0.0005
3Sources	6	1.0000	1.0000	0.0000	0.0000
BBCSport	5	1.0000	1.0000	0.0000	0.0000
Cora	7	1.0000	1.0000	0.0000	0.0000
CiteSeer	6	1.0000	1.0000	0.0000	-0.0003

Results. One estimate over the twelve datasets of the table, screen and validation pooled: **+0.0028 [-0.0046, 0.0102]**, 16/60 seed-level wins. The upper limit sits at a fifth of the +0.02 minimum useful effect, so this is a decided negative rather than an inconclusive one, and the screen and validation halves are not reported separately because neither reaches a different verdict. Two structural facts explain why. First, the measurement has no headroom on most of the suite: 89 of the 120 teacher rows exceed 0.99 fidelity, and on 3Sources, BBCSport, Cora and CiteSeer both students hit 1.0000 to four decimals. Second, the only dataset with real headroom is 100Leaves, whose 100-class target makes the regression genuinely hard (0.60 versus 0.56); the GNN does win there by +0.0392, and that single dataset is what lifts the validation mean. That fidelity win converts to only +0.0066 train AMI, so even where message passing demonstrably fits the teacher better, the extra fidelity buys no usable cluster structure.

Verdict. Dead end, and dead in the strong direction. The finding is not "the GNN failed to distil"; it is that a feature-only MLP already reproduces the fixed MDT embedding to within rotation on ten of twelve datasets, leaving nothing for message passing to recover. A3 corroborates this from the capacity side: destroying the graph outright costs the student only +0.0029 fidelity [-0.0014, 0.0072], so the neighbourhoods were carrying no fidelity to begin with. A distillation-fidelity argument for a GNN cannot be rescued by more seeds or wider hidden layers, because the baseline is at the ceiling. It could only be reopened on targets with the shape of 100Leaves — high embedding rank relative to sample count — and even there the AMI consequence would have to be demonstrated separately.

G2 — graph message passing improves MDT out-of-sample extension

Methodology. Three inductive maps are compared on identical splits, graphs and KMeans protocol. (i) `mdt_nystrom`, training-free: test rows are attached to train columns by a directed row-normalised Gaussian kNN bipartite graph (`knn_transitions`, `closure.py:128-134`), the MDT trajectory is replayed with its final step replaced by that bipartite operator (`mdt_oos_operator`), and the result is projected onto the train right singular vectors (`closure.py:425`). (ii) `teacher_mlp`, parametric and feature-only: at inference the map is `x_test → z`, with no graph involved. (iii) `teacher_gnn`, parametric and graph-aware: test hidden states additionally aggregate over the same bipartite graph, so an unseen point borrows the hidden states of its train neighbours (`forward_test`, `closure.py:267-287`). Both parametric maps distil the same fixed SVD target, so any difference between them is attributable to message passing alone.

Two accuracy metrics are reported. `test_ami` refits KMeans on the test embedding, which measures only whether the geometry is clusterable. `inductive_ami` assigns test points to centroids fitted on the train embedding (`closure.py:394-399`), which additionally requires the OOS map to land in the same frame as the train embedding; it is the stricter and more deployment-like metric. The decisive quantity is the decomposition

```
(GNN − Nyström) = (MLP − Nyström) + (GNN − MLP)
```

whose first term isolates parametric distillation and whose second term isolates the graph. G2 is a claim about the second term only.

Evidence. Per-dataset means over five seeds; `nys`, `mlp`, `gnn` are absolute AMI, not differences.

dataset	test nys	test mlp	test gnn	induct nys	induct mlp	induct gnn
MSRC-v5	0.6202	0.6892	0.6789	0.5989	0.6176	0.6726
Handwritten	0.8698	0.8756	0.8823	0.8715	0.8792	0.8835
Wikipedia	0.2923	0.2955	0.3458	0.2768	0.2783	0.2958
UCI	0.7818	0.8242	0.8283	0.8047	0.8275	0.8305
OutdoorScene	0.5232	0.5352	0.5495	0.4999	0.5291	0.5470
Caltech101-7	0.5283	0.5404	0.5459	0.5050	0.5176	0.5367
100Leaves	0.6581	0.8591	0.8849	0.6375	0.8618	0.8758
Yale	0.5398	0.4698	0.4895	0.5144	0.4523	0.5004
3Sources	0.0799	0.3535	0.2093	0.0755	0.1316	0.0776
BBCSport	0.1670	0.2231	0.2026	0.1210	0.1143	0.1303
Cora	0.0266	0.0246	0.0256	0.0000	0.0000	0.0000
CiteSeer	0.0258	0.0576	0.0466	0.0082	0.0075	0.0089

Paired effects for the decomposition:

contrast	metric	screen	validation	pooled (12)
GNN − Nyström	test	+0.0359 [0.0153, 0.0564]	+0.0602 [-0.0454, 0.1659]	+0.0480 [0.0034, 0.0927]
GNN − Nyström	inductive	+0.0349 [0.0113, 0.0584]	+0.0394 [-0.0632, 0.1420]	+0.0372 [-0.0058, 0.0801]
MLP − Nyström	test	+0.0241 [-0.0033, 0.0515]	+0.0817 [-0.0545, 0.2180]	+0.0529 [-0.0069, 0.1128]
MLP − Nyström	inductive	+0.0154 [0.0047, 0.0261]	+0.0352 [-0.0697, 0.1401]	+0.0253 [-0.0182, 0.0688]
GNN − MLP	test	+0.0118 [-0.0097, 0.0333]	-0.0215 [-0.0872, 0.0442]	**-0.0049 [-0.0352, 0.0254]**
GNN − MLP	inductive	+0.0195 [-0.0003, 0.0392]	+0.0042 [-0.0308, 0.0393]	**+0.0119 [-0.0053, 0.0290]**

Results. The composite contrast against Nyström is the only one that ever clears the threshold, and the decomposition shows it is not a graph effect. On the pooled twelve datasets the parametric term carries +0.0529 test AMI while the graph term is -0.0049, an interval that excludes the minimum useful effect in both directions of interest. The inductive term, +0.0119 [-0.0053, 0.0290], is the most favourable number message passing produces anywhere in the suite and it still fails the rule. Sign instability across datasets is severe rather than incidental: the graph helps most on Wikipedia (+0.0503 test) and hurts most on 3Sources (-0.1442 test), and the screen-to-validation flip of the mean from +0.0118 to -0.0215 is driven by that single dataset. Note also that `inductive_ami` is uniformly lower than `test_ami` for every method, confirming that part of the apparent OOS quality in the paper-compatible metric comes from refitting KMeans on test rather than from the map itself.

Verdict. G2 is a dead end. Message passing is not the source of the OOS gain, and the surviving positive result — parametric distillation beating Nyström — belongs to the MLP/DDM line, where it should be pursued without a graph. Three limitations bound how far even this negative generalises, and all three point at the same missing ingredient. First, every graph in the suite is built by kNN on the features themselves (`build_graphs` runs unconditionally), so it encodes no information the encoder cannot already read; the experiment never tested relational side information. Second, Cora and CiteSeer ship genuine citation structure only as an n×n adjacency view, which is then standardised and re-kNN'd like any other feature block, and `cap_train: 800` subsamples them from 2708 and 3312 nodes, discarding most of their edges before the graph is even built. Third, both retain almost no recoverable cluster signal for any method — Cora's inductive AMI is exactly 0.0000 across all three maps — so they contribute two near-zero rows that tighten the interval without carrying information. A reopened G2 would need graphs that are given rather than derived, and full graphs rather than subsampled ones.

G6 — GNN computation is preferable to sparse truncated SVD

Methodology. Three timers are recorded per row. `graph_seconds` covers kNN search, Gaussian weighting, symmetrisation and row normalisation over all views; it is shared by every method and is therefore excluded from the ratio. `train_seconds` covers `svds` plus the OOS projection for `mdt_nystrom`, and the complete training loop with early stopping for the neural variants. `inference_seconds` covers the forward pass over the test split. The hypothesis allows two routes to a positive verdict: an end-to-end wall-clock win, or a realistic repeated-query break-even point where amortised inference recovers the training cost.

Evidence. Per-dataset means over five seeds, in seconds.

dataset	graph (shared)	svds	GNN train	MLP train	GNN inference	GNN/svds
MSRC-v5	0.090	0.015	1.231	0.830	0.0073	82.4×
Handwritten	0.510	0.251	3.305	1.296	0.0247	13.2×
Wikipedia	0.305	0.248	1.551	0.851	0.0106	6.3×
UCI	0.063	0.193	1.924	0.922	0.0145	10.0×
OutdoorScene	0.162	0.277	2.858	1.282	0.0219	10.3×
Caltech101-7	0.340	0.268	4.342	2.350	0.0366	16.2×
100Leaves	0.079	1.168	1.832	0.976	0.0129	1.6×
Yale	0.928	0.007	2.003	1.796	0.0111	268.7×
3Sources	0.287	0.011	1.603	1.374	0.0084	147.0×
BBCSport	0.316	0.046	2.210	1.745	0.0164	47.7×
Cora	0.674	0.249	5.304	3.696	0.0476	21.3×
CiteSeer	0.498	0.210	3.584	2.834	0.0340	17.1×

Results. Training is 12.1× slower than `svds` on the screen (2.535 s versus 0.209 s) and 9.8× slower on validation (2.756 s versus 0.282 s). Including the shared graph build, end-to-end fitting is 6.1× slower on the screen and 4.3× slower on validation, 5.0× pooled. The ratio is extremely heterogeneous (1.6× on 100Leaves, 268.7× on Yale) because it is dominated by how cheap `svds` happens to be: the SVD scales with sample count and requested rank, so it is near-free on the 115-to-380-sample datasets and costliest on 100Leaves, where a rank-100 decomposition of 800 points takes 1.168 s. 100Leaves is the only case where the two are within one order of magnitude, and even there the GNN loses. The neural cost is not a message-passing artefact: `teacher_mlp` is itself 6.8× slower than `svds` pooled, so the deficit is the training loop, not the aggregation.

Verdict. Dead end on the end-to-end route, and the break-even route was never actually evaluated — a gap that should be recorded rather than glossed. `mdt_nystrom` writes `inference_seconds: 0.0` unconditionally (`closure.py:430`); its OOS projection cost is folded into `train_seconds` and never separately timed, so no amortisation curve can be drawn from these result files. The break-even argument nevertheless fails on structure rather than on missing measurement. Message passing requires the test-to-train bipartite graph at inference time (`forward_test` consumes `test_graphs`), so a GNN query still pays the kNN search against the training set — exactly the cost Nyström pays. The GNN therefore removes no term from the query path while adding a 2.5-to-2.8 s fit; its ~0.02 s forward pass cannot amortise against a baseline it has not made cheaper. Only `teacher_mlp` drops the graph at inference and so is the only variant for which a break-even question is even well posed. If that question is ever pursued, it belongs to the MLP/DDM line and requires instrumenting Nyström inference first.

Amendment A3 — the capacity-matched null for message passing

Why this was run at all. `### The single difference` argues that the zero-input MLP makes every message-passing contrast a message-passing contrast rather than a capacity contrast. That holds for nominal parameter count and fails for effective capacity: in the `*_mlp` variants the neighbour half of `layer1` and `layer2` multiplies zeros, so those weights compute no function. Any GNN win over that control therefore still admits the reading "the extra live parameters helped, not the graph". The control cannot be fixed by matching capacity, because the hop under test is the capacity. A3 attacks from the other side: keep every weight live and destroy only the correspondence between neighbourhoods and features.

Methodology. `shuffle_graphs` (`closure.py:167-181`) draws one permutation of the training node set per split and relabels every graph — `P_v[order][:, order]` on train, `B_v[:, order]` out of sample — while the features stay in place. Preserved exactly: row stochasticity, `nnz`, `k`, the multiset of degrees, the kernel weight values, and the spectrum of each `P_v` (relabelling is a similarity transform by a permutation matrix). Preserved in the model: all weights active, `message_passing` true, since the selector is `endswith("mlp")`. Destroyed: which node's features arrive at which node. A single permutation is shared across views, matching the real setting where every view indexes the same node set. `teacher_shuffled` and `gae_shuffled` are dispatched in `run_one` (`:498-500`) and are otherwise identical to `teacher_gnn` and `gae_uniform`: same seed, splits, bandwidths, `k`, trajectories, epoch budget, patience, KMeans protocol, frozen uniform fusion.

The GAE arm needs one extra care. Its reconstruction target is the union of the true per-view kNN graphs, so `fit_model` takes a separate `edge_graphs` argument (`:355`, `:373`) and only the aggregation graphs are shuffled. Shuffling the target as well would make it unlearnable, and that collapse would be misread as evidence for message passing.

Grids. Both were produced by the code as it stood before the α parameter existed, which is the plain row-stochastic construction the screen used; the rows carry no `alpha` field. Teacher and Nyström rows reproduce `results/gnn_mdt/metrics.jsonl` to 0.0 maximum absolute difference, so the shuffled machinery is inert on the unshuffled arms.

- Control grid: the six screen datasets × five seeds, 360 rows, `results/gnn_mdt/control_metrics.jsonl` — directly comparable to the locked run.
- Power grid: twelve datasets, 612 rows, `results/gnn_mdt/power_metrics.jsonl`, summary rebuilt into `power_summary.json`.

The twelve are the six screen datasets plus 3Sources, BBCSport, Prokaryotic, Reuters-1200, NUS-WIDE and MNIST-4. **This is not the ledger's pooled twelve.** 3Sources and BBCSport belong to the untouched-validation six and are already spent; Prokaryotic, Reuters-1200, NUS-WIDE and MNIST-4 are new to this ledger; 100Leaves, Yale, Cora and CiteSeer are absent. Those four were excluded before the run on stated grounds — Cora and CiteSeer carry an adjacency-as-feature view whose columns misalign once `cap_train` subsamples the rows, and 100Leaves puts 100 classes against 800 training points — but the ledger's own rule retained them rather than dropping them post hoc. The A3 mix is therefore the more favourable one and cannot overturn a validation-set closure. It is a power and control analysis, not a fresh validation.

Evidence. Per-dataset means on the power grid; GAE columns are train AMI, teacher columns are test AMI.

dataset	gae unif	gae mlp	gae shuf	Δ mp	Δ signal	teacher gnn	teacher shuf	Δ signal
3Sources	0.0712	0.0584	0.0550	+0.0128	+0.0162	0.2184	0.2818	-0.0634
BBCSport	0.1965	0.1121	0.1240	+0.0844	+0.0724	0.1868	0.2374	-0.0506
Caltech101-7	0.5750	0.5161	0.5469	+0.0589	+0.0281	0.5565	0.5510	+0.0054
Handwritten	0.8511	0.7960	0.7810	+0.0551	+0.0700	0.8823	0.8768	+0.0055
MNIST-4	0.7850	0.7841	0.7519	+0.0008	+0.0331	0.7071	0.7076	-0.0005
MSRC-v5	0.6731	0.6776	0.6590	-0.0045	+0.0140	0.6789	0.7155	-0.0365
NUS-WIDE	0.1196	0.1157	0.1151	+0.0039	+0.0045	0.1092	0.1126	-0.0034
OutdoorScene	0.5698	0.5445	0.5204	+0.0254	+0.0494	0.5520	0.5480	+0.0040
Prokaryotic	0.2219	0.1684	0.1310	+0.0535	+0.0909	0.0812	0.0385	+0.0427
Reuters-1200	0.1522	0.1174	0.1166	+0.0348	+0.0356	0.0931	0.1021	-0.0090
UCI	0.8465	0.8132	0.7882	+0.0334	+0.0584	0.8245	0.8138	+0.0106
Wikipedia	0.3928	0.3630	0.2908	+0.0298	+0.1020	0.3458	0.3057	+0.0400

Paired effects with the ledger's estimator: seeds averaged within a dataset, then a paired 95% Student interval across the dataset means.

contrast	metric	control, 6 datasets	power, 12 datasets
`gae_uniform − gae_shuffled`	train	+0.0543 [+0.0199, 0.0887], 26/30	+0.0479 [+0.0280, 0.0678], 44/51
`gae_uniform − gae_shuffled`	test	+0.0603 [+0.0326, 0.0879]	+0.0559 [+0.0300, 0.0817]
`gae_uniform − gae_shuffled`	fidelity	+0.1641 [+0.0580, 0.2702]	+0.0949 [+0.0361, 0.1536]
`gae_uniform − gae_mlp`	train	+0.0303 [+0.0083, 0.0523], 25/30	+0.0324 [+0.0152, 0.0495], 40/51
`gae_shuffled − gae_mlp`	train	-0.0240 [-0.0584, 0.0103], 12/30	-0.0155 [-0.0323, 0.0012], 23/51
`gae_shuffled − gae_mlp`	test	-0.0189 [-0.0397, 0.0018]	-0.0317 [-0.0576, -0.0058]
`teacher_gnn − teacher_shuffled`	test	+0.0049 [-0.0208, 0.0307], 16/30	-0.0046 [-0.0250, 0.0158], 24/51
`teacher_gnn − teacher_shuffled`	inductive	+0.0147 [-0.0085, 0.0378]	+0.0109 [-0.0012, 0.0230]
`gae_uniform − mdt_consensus`	test	+0.0168 [-0.0422, 0.0758]	+0.0340 [-0.0089, 0.0769]
`gae_learned − mdt_consensus`	test	+0.0227 [-0.0387, 0.0840]	+0.0328 [-0.0120, 0.0776]

Results. The capacity reading of the GAE effect is eliminated. `gae_shuffled` runs the full architecture with every weight live and loses 0.0479 train AMI [+0.0280, 0.0678] to the identical model on true graphs, on 44 of 51 pairs. Had the `gae_uniform − gae_mlp` gain come from the extra live parameters, the shuffled arm would have matched `gae_uniform`; it does not come close. The same contrast on Procrustes fidelity, +0.0949 [+0.0361, 0.1536], shows the structure the GAE exploits is MDT-aligned rather than incidental.

The stronger reading — wrong neighbours are worse than no neighbours — is directionally consistent everywhere but established on only one metric. The mean ordering puts `gae_shuffled` last on both grids, and the contrast against `gae_mlp` excludes zero on test AMI at twelve datasets (-0.0317 [-0.0576, -0.0058]) while crossing zero on train AMI on both grids (-0.0240 [-0.0584, 0.0103] and -0.0155 [-0.0323, 0.0012]). Report it as a tendency, not a result. The load-bearing claim does not depend on it.

On the teacher arm the graph is worth nothing, and A3 upgrades that from a failure to replicate into a positive null: destroying the graph entirely costs -0.0046 test AMI [-0.0250, 0.0158] over twelve datasets, an upper limit below the +0.02 minimum useful effect, hence dead end under the suite's own rule. The mechanism is the one G1 already identified from the other direction — the regression target is the MDT embedding, so the graph is already inside the label and message passing has nothing left to add. The inductive metric is the one place a residue survives, +0.0109 [-0.0012, 0.0230], which is the same sub-threshold hint G2 recorded and is not enough to reopen anything.

`gae_vs_consensus` does not resolve, and not for want of power. The effect is stable across the two grids (+0.0227 → +0.0328 learned; +0.0168 → +0.0340 uniform) while the half-width fell from 0.061 to 0.045, so it sits on the +0.02 threshold rather than drifting toward it. Compute is 14.1× the sparse SVD on the power grid. A real but threshold-sized gain at fourteen times the cost is what the minimum-effect rule exists to reject.

Reproducibility. Teacher, Nyström and consensus rows are bit-exact across identical invocations. GAE rows are not: the same dataset/seed/method cell moves by up to 0.0205 train AMI between two runs of the same six datasets. The cause is early stopping on `value < best_loss - 1e-6` applied to a loss whose sparse matrix products are not bit-reproducible, so an infinitesimal difference shifts the stopping epoch and the restored best state. Aggregates absorb it — the `gae_uniform` − `gae_mlp` effect moved 0.0315 → 0.0303 between the two runs of the screen six — so cite GAE aggregates and never a single GAE row.

Completeness. The power grid is 612 of 720 rows: seeds 0-3 finished on all twelve datasets, seed 4 only on MSRC-v5, Handwritten and Wikipedia, because the process was killed rather than failing. Nine of the twelve dataset means therefore average four seeds instead of five. `--resume` would finish it; given that every effect above is stable from six to twelve datasets, no verdict is expected to move.

Verdict. A3 succeeds as a control and reopens nothing. It removes one specific escape route — "the GNN only won because more of its weights were alive" — and in doing so converts the GAE message-passing gain into a demonstrated graph-structure effect on this dataset mix, and the teacher arm's graph term into a demonstrated null. Neither changes a decision: G2 stays a dead end and is now dead for a stated reason rather than for a missing replication, and G3 stays closed because a threshold-sized advantage over `mdt_consensus` at 14× the compute still fails the useful-effect test. The lasting instruction is procedural: a zero-input ablation is not a capacity control on its own, and any future message-passing claim in this line should ship its shuffled-graph arm in the same run.

Amendment A2 — the AlphaFold family on the OOS map

Scope, and what was deliberately not built. The GNN closure leaves one live result: parametric out-of-sample extension by a feature-only map. The question here is whether the DeepMind Alpha-family design ideas improve that map. Only two of them survive contact with this problem, and the ledger's own numbers are the reason:

- **Recycling** (AF2's iterative refinement) is tested. It attacks the observed failure mode directly — `inductive_ami` is below `test_ami` for every method in the suite, meaning the OOS map lands in a slightly wrong frame — and it adds almost no parameters, so a gain could not be a capacity artefact.
- **Pair-representation targets** (AF predicts pair geometry rather than coordinates) are tested. This is the one place where the Alpha-family framing says something structural about MDT: the student currently regresses `U_k Σ_k`, whose frame is arbitrary, which is exactly why `procrustes_fidelity` has to rotate before measuring. A Gram target removes that gauge freedom.
- **Evoformer stacks, triangle multiplicative updates, axial attention, MSA and template modules are not built.** Templates are retrieval over a training set and are structurally what `mdt_nystrom` and message passing already do, closed under G2. The rest is capacity, and capacity is not the binding constraint on a rank-k SVD target that `teacher_mlp` already reproduces at 0.9904 mean Procrustes fidelity with 89 of 120 rows above 0.99. Adding an attention stack here would repeat the confound the shuffled-graph null exists to catch.

Methodology. A 2×2 factorial over {coordinate, pair} × {no recycling, 3 cycles}, all feature-only, all distilling the same fixed MDT SVD teacher, on the six screen datasets × five seeds; 210 rows in `results/gnn_mdt/af_metrics.jsonl`, verdicts in `results/gnn_mdt/af_summary.json`. Recycling re-enters the previous cycle's embedding into the fused input through a projection that is **zero-initialised** (`MultiViewSAGE.__init__`, `closure.py:276-281`, applied at `:303` and `:319-321`), so an untrained recycled model is bit-identical to `teacher_mlp` and the baseline is nested inside the candidate; all cycles but the last run under `no_grad`, as in AF2, so the added cost is forward-only (`closure.py:410-416`, inference loop `:447-453`). The pair arm replaces `MSE(z, F)` with `MSE(z zᵀ, G)` where `G = F Fᵀ` scaled to unit RMS (`closure.py:395-396`, loss at `:418`); it adds no parameters at all. Variant flags are substring tests, so `teacher_mlp_pair_recycle` composes both (`:377-380`), and `recycles` defaults to 3 (`:545`). Everything else is untouched: same splits, seeds, graphs, epoch budget, patience, optimiser and KMeans protocol, `inductive_ami` pre-declared as primary because it is the metric that punishes landing in the wrong frame.

Evidence. Per-dataset means over five seeds. `mlp` is the current survivor, `nys` the training-free baseline.

dataset	induct nys	induct mlp	+ recycle	+ pair	+ pair & recycle
MSRC-v5	0.5989	0.6176	0.6231	0.6578	0.6448
Handwritten	0.8715	0.8792	0.8793	0.8594	0.8583
Wikipedia	0.2768	0.2783	0.2647	0.3014	0.3041
UCI	0.8047	0.8275	0.8312	0.8045	0.8084
OutdoorScene	0.4999	0.5291	0.5266	0.5224	0.5368
Caltech101-7	0.5050	0.5176	0.5199	0.5216	0.5350

dataset	test nys	test mlp	+ recycle	+ pair	+ pair & recycle
MSRC-v5	0.6202	0.6892	0.6554	0.6999	0.6932
Handwritten	0.8698	0.8756	0.8855	0.8579	0.8661
Wikipedia	0.2923	0.2955	0.2921	0.3178	0.3081
UCI	0.7818	0.8242	0.8231	0.8148	0.8114
OutdoorScene	0.5232	0.5352	0.5331	0.5488	0.5501
Caltech101-7	0.5283	0.5404	0.5413	0.5440	0.5428

Paired effects, seeds averaged within a dataset then a paired 95% Student interval across the six dataset means:

arm	metric	vs `teacher_mlp`	vs `mdt_nystrom`
recycle	inductive	-0.0008 [-0.0080, 0.0064]	+0.0147 [-0.0012, 0.0305]
recycle	test	-0.0049 [-0.0206, 0.0107]	+0.0192 [+0.0025, 0.0358]
recycle	train	-0.0032 [-0.0075, 0.0011], 6/30 wins	+0.0062 [+0.0005, 0.0120]
pair	inductive	+0.0029 [-0.0231, 0.0290]	+0.0184 [-0.0072, 0.0440]
pair	test	+0.0038 [-0.0119, 0.0196]	+0.0279 [-0.0034, 0.0592]
pair & recycle	inductive	+0.0063 [-0.0163, 0.0290]	+0.0218 [-0.0015, 0.0450]
pair & recycle	test	+0.0019 [-0.0099, 0.0138]	+0.0260 [-0.0011, 0.0531]
pair	fidelity	**-0.0850 [-0.1395, -0.0305]**, 0/30 wins	-0.0945 [-0.1550, -0.0340]
recycle	fidelity	+0.0003 [-0.0000, 0.0006]	-0.0093 [-0.0205, 0.0019]

Results. A2a is a decided negative: recycling's inductive effect is -0.0008 with an upper limit of 0.0064, a third of the minimum useful effect, so the rule labels it dead end rather than inconclusive. It is mildly harmful where it is measurable at all — train AMI -0.0032 with

6/30 wins, test AMI -0.0049 — and it is not free: training goes from 1.001 s to 1.623 s and inference from 6.93 ms to 20.11 ms, a 2.9× query cost for a negative effect. The mechanism is not mysterious. AF2 recycling refines a structure prediction against a geometry module that can be re-evaluated; here the student regresses a fixed linear-algebraic target in one shot, so the second and third passes have nothing new to condition on and the recycled input acts as noise the model must learn to ignore — which, given the zero-initialised projection, is exactly what it mostly does.

A2b does not clear the bar either: +0.0029 inductive, +0.0038 test, both intervals straddling zero, and the combination with recycling adds nothing beyond the pair target alone (+0.0063 inductive), so there is no interaction to exploit. Its dataset heterogeneity is severe and structured: the pair target gains on the two datasets where the coordinate map is weakest relative to Nyström (MSRC-v5 +0.0402, Wikipedia +0.0231 inductive) and loses on the two where the coordinate map is already strong (Handwritten -0.0198, UCI -0.0230). That is a real pattern, but it is a per-dataset trade, not an improvement.

The one unambiguous finding is a negative that matters more than either hypothesis. The pair arm loses **0.0850 Procrustes fidelity** to the teacher, 0/30 wins, while its AMI is statistically indistinguishable from the coordinate arm's. MSE over the Gram weights each direction by roughly the square of its singular value, so the pair arm fits the dominant directions and neglects low-variance ones, which column-standardised Procrustes counts equally — hence fidelity collapses to 0.86 on Handwritten and 0.85 on UCI and Wikipedia while clustering quality does not move. G1 already showed that a fidelity advantage buys no AMI; A2b shows the converse, that a large fidelity loss costs no AMI either. Procrustes fidelity to the fixed MDT target is therefore not a proxy for anything the suite cares about, in either direction, and should be reported as a distillation diagnostic only.

Verdict. Both arms closed on the screen, so no untouched-data validation was spent. Recycling is a dead end with a cost attached; the pair target is a dataset-dependent trade with no mean gain and a large fidelity penalty. The Alpha-family transplant does not improve MDT out-of-sample extension, and the best arm against the training-free baseline (+0.0218 inductive for pair & recycle) is not better than the plain feature-only map that was already the survivor. The honest reading is that this suite has now tested three different mechanisms on top of parametric OOS distillation — message passing, iterative refinement, and gauge-free targets — and none of them beats the two-layer feature-only map, which is consistent with the theorem the ledger opens with: at the global optimum the target is a rank-k factorisation, and the remaining headroom is in what target is chosen, not in the machinery that fits it. A reopened Alpha-family branch would need to change the teacher — for instance a target that is itself refined across cycles, which is what AF2 actually does and what none of these arms did.

Literature anchors

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